

MTHATHA, South Africa

WMO: 686680

Lat: 31.5497S

Lon: 28.6739E

Elev: 2451

StdP: 13.44

Time Zone: 2.00 (E02)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	37.3	39.4	19.2	15.8	65.9	23.1	19.0	65.2	25.2	57.9	21.7	61.5	2.3	270	0.399

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	18.3	94.2	67.7	91.0	67.3	87.7	66.9	73.9	85.5	72.3	83.7	70.9	81.7	8.4	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
70.5	123.0	79.6	68.8	116.0	77.9	67.6	111.2	76.3	39.3	85.5	37.7	83.7	36.4	82.0	90.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
20.9	17.7	15.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
			DB												
			WB	31.8	77.4	1.0	1.3	31.0	78.4	30.4	79.1	29.8	79.9	29.1	80.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	64.6	71.6	72.1	70.0	65.7	61.9	56.3	56.1	59.5	62.2	64.7	66.3	69.0
	DBStd	7.90	5.42	5.29	5.72	6.05	5.53	4.72	5.21	6.12	6.71	6.78	6.67	5.83
	HDD50	20	0	0	0	0	1	6	9	3	1	0	0	0
	HDD65	1268	10	6	19	65	126	262	278	192	130	92	61	27
	CDD50	5340	669	619	620	470	370	196	199	298	368	454	489	588
	CDD65	1115	214	206	174	85	30	2	3	22	47	81	101	150
	CDH74	9056	1389	1326	1164	712	396	105	158	418	650	809	850	1079
	CDH80	3372	531	501	448	241	88	11	20	134	264	335	364	435
Wind	WSAvg	6.2	7.1	6.5	5.9	5.0	4.5	4.7	4.9	5.9	6.7	7.2	7.9	7.6
Precipitation	PrecAvg	25.0	3.5	3.6	3.9	2.0	0.8	0.5	0.6	0.8	1.1	2.6	3.0	3.4
	PrecMax	36.2	6.4	7.8	10.3	9.6	4.1	3.5	3.1	3.9	6.3	10.7	8.1	8.7
	PrecMin	14.9	0.4	0.5	0.5	0.4	0.0	0.0	0.0	0.0	0.4	0.1	0.6	0.6
	PrecStd	5.3	1.6	1.9	2.4	1.7	0.9	0.8	0.7	0.8	1.1	1.9	1.6	1.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	96.8	96.5	95.2	91.5	86.4	81.1	82.6	90.0	93.3	95.1	96.8	96.1
		MCWB	70.4	71.2	69.5	66.2	61.0	56.0	54.7	58.3	59.9	63.7	64.1	70.0
	2%	DB	91.3	90.5	89.9	86.7	82.5	77.2	78.7	83.9	87.4	89.5	89.8	90.0
		MCWB	70.5	71.4	69.3	65.0	59.1	54.1	53.6	55.9	58.8	62.9	67.2	69.2
	5%	DB	86.7	85.8	85.6	82.4	78.8	73.4	74.6	78.8	81.8	83.9	83.7	85.0
		MCWB	70.8	71.2	68.7	63.3	58.1	53.0	52.7	55.1	58.7	63.3	66.5	69.1
	10%	DB	82.4	81.9	80.9	77.6	74.3	69.4	69.9	73.3	75.6	78.3	78.1	80.1
		MCWB	69.9	70.2	68.1	63.3	57.9	51.8	51.5	53.8	58.0	62.8	64.6	67.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	75.3	76.5	75.9	70.9	67.3	58.5	58.4	61.8	65.3	69.5	71.8	74.3
		MCDB	88.5	87.5	85.8	80.7	74.9	72.9	71.9	74.6	80.4	83.5	84.6	86.7
	2%	WB	73.1	73.7	72.6	67.9	62.6	56.5	55.9	59.7	62.9	66.6	69.3	71.8
		MCDB	85.8	84.4	83.8	80.0	72.4	70.3	69.1	72.9	76.8	80.0	81.7	84.2
	5%	WB	71.7	71.9	70.3	66.2	60.8	55.1	54.4	58.1	61.1	64.9	67.5	70.0
		MCDB	83.3	82.3	80.3	77.7	72.4	67.9	68.0	70.7	74.1	78.1	79.3	81.8
	10%	WB	70.1	70.5	68.7	64.7	59.2	53.5	53.0	56.2	59.5	63.2	65.5	68.3
		MCDB	80.3	80.2	77.8	74.6	70.5	64.9	66.6	67.7	70.8	75.2	75.7	78.1
Mean Daily Temperature Range	5% DB	MDBR	18.3	18.3	18.6	21.4	24.2	26.0	27.3	25.8	24.0	21.5	19.5	18.9
		MCDBR	26.7	25.7	27.4	29.1	31.0	32.9	34.0	34.5	34.8	32.6	30.2	29.4
		MCWBR	11.0	11.1	11.3	10.9	11.6	13.4	13.4	12.6	12.0	12.3	11.4	12.0
	5% WB	MCDBR	23.4	22.7	23.0	24.0	24.2	27.3	28.8	26.6	26.5	25.8	24.6	24.7
MCWBR		10.7	10.9	10.8	10.4	11.2	12.4	13.1	11.4	11.5	11.6	11.3	11.7	
Clear-Sky Solar Irradiance	taub	0.385	0.387	0.369	0.335	0.312	0.293	0.299	0.357	0.397	0.402	0.378	0.383	
	taud	2.393	2.392	2.414	2.469	2.497	2.514	2.485	2.331	2.227	2.262	2.372	2.392	
	Ebn at Noon	303	296	290	284	275	274	277	272	276	288	303	305	
	Edh at Noon	40	39	36	30	26	24	26	34	43	44	41	41	
All-Sky Solar Radiation	RadAvg	1953	1812	1582	1323	1097	977	1062	1268	1542	1710	1906	1948	
	RadStd	141	153	107	76	57	62	62	80	103	153	133	128	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (6 neighbors)	+0.43	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page